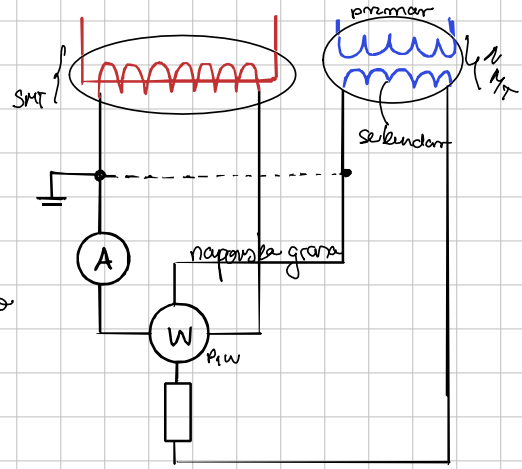
Mjerenje snage u trofaznim sustavima [4]

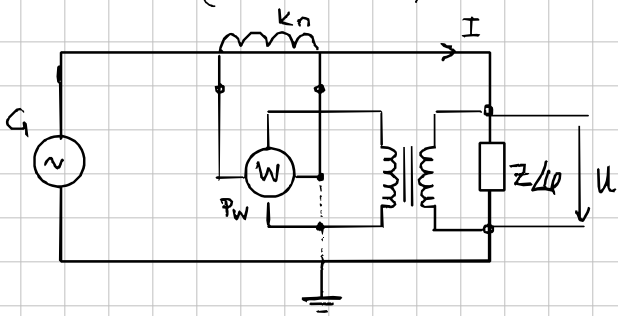
Neizravni načini mjerenja snage P



- kod viših napona => SMT + NMT

=> mjerenu snagu dobivamo tako da očitanje na vatmetrima pomnožimo s prijenosnim omjerom SMT-a i NMT-a

$$P = K_I \cdot K_U (P_{1W} + \dots + P_{nW})$$



- kada su struja i napon preveliki, => smanjimo ih transformat.
- mjerimo struju i napon sekundara i umnožimo faktorom prijenosa

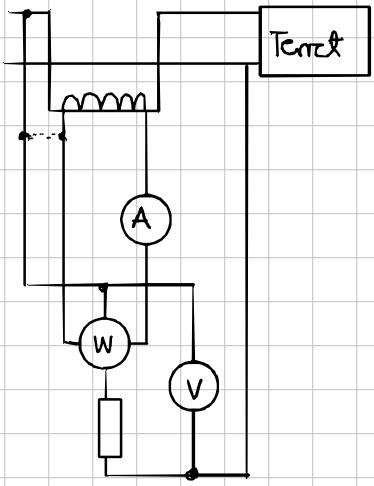
$$P = K_U \cdot K_I \cdot P_W = U \cdot I \cdot \cos(\varphi + \delta_u - \delta_i)$$

→ transformatori unose amplitudne i fazne pogreške

$$\Rightarrow \varepsilon_p = -(\delta_u - \delta_i) \operatorname{tg}\varphi \cdot 100\%$$

Poluizravno mjerenje snage

- kad se premaši granica izravnog mjerenja veličina (napon / struju)
  - ↳ uključuje SMT / NMT
    - ↳ smanjujemo struju / napon
  - ↳ struja: 50 A
    napon: 1000 V

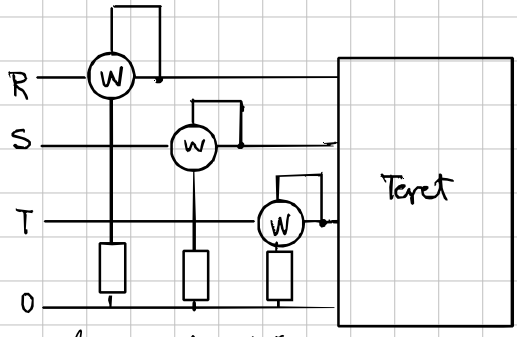


- napon 750 V → SMT, strujna grana spaja na sekundar

$$P = P_W \cdot K_I$$

Mjerenje snage P metodom 3 vatmetra

- Četverožični sustav: 3 faze (R, S, T) i nulvodič
- strujne grane vatmetra = FAZNE STRUJE
- naponske grane = FAZNI NAPONI
  ↳ svaki vatmetar mjeri snagu jedne faze

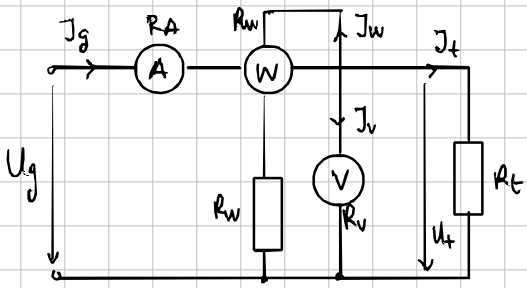


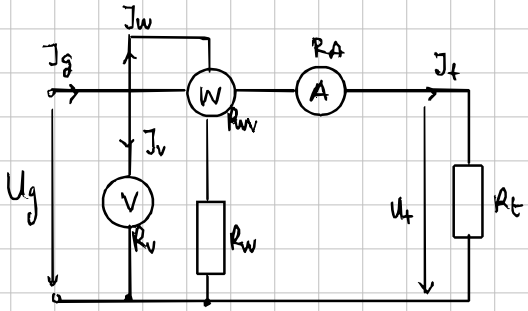
$P = P_{WR} + P_{WS} + P_{WT}$ → moguće i bez nul vodiča

Izravno mjerenje snage

Mjerenje snage pomoću W, V, A

- Sami vatmetri ne mjere samu snagu, već i vlastite gubitke nastale zbog unutarnjeg otpora njegovih grana
- ampermetrom i voltmetrom se kontrolira opterećenje strujne (otpor R_W) i naponske grane vatmetra (otpor R_V)
- radi se kod istosmjerne i kod izmjenične struje





$$P_t = P_W - \left(\frac{U_t^2}{R_V} + \frac{U_t^2}{R_W}\right)$$

$$P_t = P_W - J_t^2 (R_A + R_W)$$

$$P_g = P_W + J_g^2 (R_A + R_W)$$

$$P_g = P_W + \frac{U_g^2}{R_V} + \frac{U_g^2}{R_W}$$

↳ ukupna snaga koju isporučuje generator

⇒ ovim spojem može se odrediti i $\cos\varphi$ trošila

- uvjet: struja J_W kroz naponsku granu mora biti u fazi s naponom na naponskoj grani

$U_t = U_V$

$$J_t = J_A - \frac{U_V}{R_W} - \frac{U_V}{R_A}$$

$$P_t = P_W - \left(\frac{U_V^2}{R_V} + \frac{U_V^2}{R_W}\right)$$

$$\cos\varphi_t = \frac{P_t}{U_t \cdot J_t} = \frac{P_W - \left(\frac{U_V^2}{R_V} + \frac{U_V^2}{R_W}\right)}{U_V\left(J_A - \frac{U_V}{R_W} - \frac{U_V}{R_A}\right)}$$